25/11/16

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR

B.E. 3rd SEMESTER (CS) FINAL EXAMINATION, 2016

Digital logic (CS301)

FULL MARKS: 70

TIME: 3 Hrs

(Answer any five)

1. (a) Find the 9's complement of the following 8-digit decimal numbers.
(i)12349876 (ii) 00980100 (iii) 90009951 (iv) 67543218
(b) Perform subtraction with the following unsigned decimal numbers by taking the 10's complement of the subtrahend.
(i) 5250-1321 (ii) 1753-8640 (iii) 20-100 (iv) 1200-250
(c) In a DTL circuit voltage across conducting diode is 0.7 volt, cut in voltage of the diode is 0.6 volt, cut in voltage of the transistor is 0.5 volt, V_{CEsat}=0.2 volt, V_{BEsat}=0.8 volt, β=30. Find the logic 0 and logic 1 noise margin.
(d) The state of a 12-bit register is 010110010111. What is the content of the register in BCD and in excess-3 code?
[2+6+4+2]

2. (a) Prove that $xy+x'z+yz = xy+x'z$
(b) Simplify the following Boolean expressions to a minimum number of literals.
(i) $x'yz+xz$ (ii) $xy+x(wz+wz')$
(c) Find the complement of $F = x+yz$. Show that $F.F'=0$ and $F+F'=1$
(d) Convert the following expressions into SOP and POS.
(i) $(AB+C)(B+C'D)$ (ii) $x'+x(x+y')(y+z')$
(e) Show that dual of Exclusive-OR is its complement.
[1+3+4+4+2]

3. (a) Simplify the following Boolean function using K-map.
(i) $F(A, B, C, D) = \prod(0, 1, 2, 3, 4, 10, 11)$
(b) Find the essential prime implicants of the following Boolean function.
$F(W, X, Y, Z) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$
(c) Simplify F together with the don't care condition.
$F(A, B, C, D) = \sum(0, 1, 2, 9, 11)$, $d(A, B, C, D)=\sum(8, 10, 14,15)$
(d) Simplify the following Boolean functions using K-map and implement them using 3-level NOR gate.
(i) $F(W, X, Y, Z)=WX'+Y'Z'+W'YZ'$ (ii) $F(W, X, Y, Z) = \sum(5, 6, 9, 10)$ [2+2+3+7]

4. (a) Design a combinational circuit with three inputs (A, B, C) and one output (Y). The output is equal to 1 when the binary value of the input is less than 3. Otherwise the output is 0.
(b) Design a combinational circuit that adds one to a 4-bit binary number ($A_3A_2A_1A_0$) using four half adders.
(c) Prove the followings:
(i) $x'\oplus y = x\oplus y' = (x\oplus y)' = xy+x'y'$ (ii) $x\oplus 1 = x'$ and $x\oplus 0 = x$ (iii) $x\oplus y = x+y$ if $xy=0$
(d) Define the following terms.
(i) Fan out (ii) Propagation delay (iii) Noise margin
[2+2+2+1+1+6]

5. (a) Implement the following Boolean function using 8x1 MUX.
$F(A, B, C, D) = \sum(0, 3, 5, 6, 8, 9, 14, 15)$
(b) Draw the block diagram of a 5x32 decoder using four 3x8 decoders with enable input and one 2x4 decoder.
(c) Design a combinational circuit that compares two 4-bit numbers A and B to check if they are equal. The output of the circuit (X) is 1 if A=B and 0 if A≠B.
(d) Draw and explain the operation of a CMOS NAND.
[4+5+2+3]

6. (a) Draw the circuit of a 4-bit binary up ripple counter using negative edge triggered J-K flip flop.
(b) A flip flop has 10 ns propagation delay. What is the maximum delay of a 10-bit binary ripple counter and synchronous counter using such flip flops?
(c) How many clock pulses are required to shift 8-bit data through a 4-bit shift register?
(d) Write the output of the circuit in figure 1 for six clock pulses. Assume that the initial value of Q is 0.

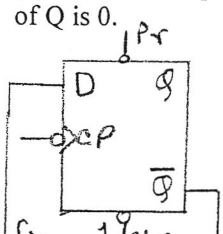

figure 1

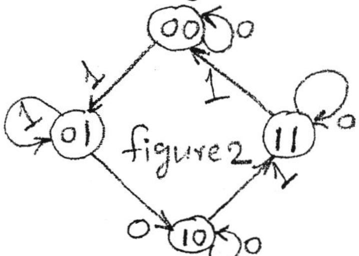

figure 2

(e) Design a clocked sequential circuit whose state diagram is shown in figure 2. Design the circuit using J-K flip flop. [3+2+1+2+6]

7. (a) What is the content of the shift register in figure 3 after 3 clock pulses? Assume that the initial content of the register is 0110.
(b) The characteristic table of X-Y flip flop is given below. Implement it using J-K flip flop.

X	Y	Q_{n+1}
0	0	1
0	1	Q_n
1	0	Q_n'
1	1	0

(c) What is the mod value of an n-bit ring counter and switch-tail counter?
(d) Draw the block diagram of a 16x4 RAM. Explain the read and write operation.
(e) Find the mod value of the counter in figure 4. [3+2+2+4+3]

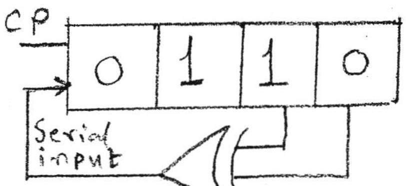

figure 3

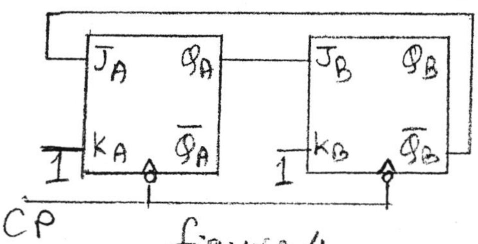

figure 4